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Surplus food commercialization in China: an attitude-behaviour-context based strategy for food waste reduction in the retail and hospitality sectors

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Abstract

Purpose – This study integrates the theory of planned behaviour and attitude-behaviour-context theory to identify determinants (price consciousness, sales proneness and food waste awareness) and barriers (perceived risks) of consumer purchase intentions toward surplus food in two different sectors in China. Specifically, it focuses on the retail sector and on the hotels, restaurants and cafés (here on HoReCa) sector. This distinction is relevant because, while HoReCa consumers prioritize trust and immediacy of consumption, retail shoppers emphasize economic incentives and autonomy.

Design/methodology/approach – An online survey was administered to a panel of consumers in China who were above 18 years old and responsible for household food purchasing. Respondents ($N = 695$) were predominantly female, educated to the graduate level, aged 31–49, and in full-time employment. Partial least squares structural equation modeling was chosen for its ability to analyse complex models and support exploratory research in emerging fields like surplus food commercialization.

Findings – Attitudes and personal norms positively influence surplus food purchase intentions in the retail and HoReCa sectors. Emphasizing economic and environmental benefits and ensuring quality and safety and foster positive attitudes. Social, psychological and financial risks do not significantly impact consumers' attitudes towards surplus food. Perceived behavioural control enhances purchase intentions in retail but not in HoReCa, likely due to differences in food availability and consumer autonomy.

Practical implications – To drive acceptance, businesses and policymakers should improve surplus food accessibility in HoReCa, enhance transparency about quality and safety and leverage economic incentives with environmental benefit. There is potential for broader adoption, making these measures crucial for reducing food waste and advancing sustainability development goals.

Originality/value – This study addresses a gap in understanding motivations and barriers towards surplus food commercialization in China, a context characterized by rapid economic growth, evolving consumer preferences and increasing urgency to address food waste in the retail and HoReCa sectors. Besides, China's unique cultural norms (e.g. tension between frugality in retail and social status in HoReCa) have been overlooked in prior studies, creating a critical gap in sector-specific strategies.

Keywords Surplus food, Food waste, Sustainability, Perceived risks, Retail and HoReCa, Consumer behaviour

Paper type Research article

1. Introduction

Food waste is a global crisis, with 1.3 billion tons discarded annually, contributing to environmental degradation, economic losses, and social inequities (Long *et al.*, 2023). Surplus food (here on SF) commercialization represents a promising solution to this issue (Derqui and Filimonau, 2024; Cakar, 2022). SF refers to safe, edible excess inventory (e.g. near-expiry

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retail items, unsold HoReCa meals), that is typically strategically priced by supermarkets, food stores or restaurants, to reduce waste while ensuring quality (Teigiserova *et al.*, 2020).

In China, food waste reached 61.37 million tons in 2020 and is projected to double by 2040 (Ogunmoroti *et al.*, 2022; Xue *et al.*, 2021). Rapid urbanization, rising incomes, and shifting consumption patterns have intensified food waste in recent years in China, particularly in urban areas (Wu *et al.*, 2023) and in the HoReCa sector (Li *et al.*, 2019). In response, the Chinese government enacted the Anti-Food Waste Law in 2021, mandating stricter oversight of food waste and management practices for approaching shelf-life items by food operators. This accelerated the interest in SF commercialization through specialized supermarkets and platforms for related stakeholders (Lyu *et al.*, 2024). This new legislation has heightened consumer awareness of food waste and also put pressure on businesses and supply chains to adopt sustainable practices, reshaping consumer purchasing behaviours and food consumption patterns (Wang *et al.*, 2024).

Given these regulatory changes and evolving consumer behaviours, exploring consumer acceptance of SF becomes particularly important. Understanding why consumers accept or reject SF adoption is critical, especially in countries such as China, given its rapidly growing population and increasing urbanization, which place immense pressure on food supply chains and resource sustainability. Additionally, the research on SF commercialization requires attention to the particularities and conditions of different industries, such as the retail and HoReCa sectors, given their contextual differences (Cao *et al.*, 2023). While retailers handle SF through discounts and focus on resale, the HoReCa sector faces higher waste due to portioning and perishability, emphasizing waste reduction strategies (Bolos *et al.*, 2022; Pandey *et al.*, 2023). Coherently, the present study aims to address the phenomenon of food waste consumption in the retail and HoReCa sectors in China by exploring how motivations and barriers shape consumer attitudes and purchase intentions toward SF in both sectors, offering insights into the evolving dynamics of SF commercialization in this country.

Despite the food waste crisis and growing interest in SF by companies, governments and academics, consumer resistance continues to be relevant, and little is known about the main reasons that explain this rejection. Existing research on this area presents different research gaps. First, current research focuses on Western countries, neglecting, for instance, China's unique cultural, regulatory, and contextual drivers and its potential to contribute to reducing food waste (Cao *et al.*, 2023). Indeed, China's collectivistic culture prioritizes social harmony and "face" (*mianzi*; reputation and social standing) (Gao *et al.*, 2021). Moreover, principles of traditional Chinese medicine, perceptions of a healthy diet, family harmony, and environmental concerns significantly influence food choices among Chinese consumers (Wang-Chen *et al.*, 2022). This contrasts sharply with Western contexts, where individual planning is emphasized over collective responsibility (Nikravech *et al.*, 2023). SF adoption reflects this duality: consumers may avoid SF for fear of appearing frugal ("cheap") yet embrace it as a sustainable choice, enhancing social esteem. This tension is amplified by China's environmental ethic rooted in traditional beliefs, which frames sustainability as a collective duty (Birkin *et al.*, 2021), and by discount-driven promotions and competitive pricing strategies that dominate the market (Adel *et al.*, 2022).

Second, little is known about how perceived risks affect consumers' attitudes and behaviours toward SF. Previous studies have mainly focused on investigating motivational factors shaping consumer behaviour toward suboptimal food mostly in retail channels (Aschemann-Witzel *et al.*, 2020) while ignoring the perceived barriers and risks associated with these products. Studies indicate that consumers often reject SF even though they recognize the importance of reducing food waste, mostly due to concerns about safety and quality (Aleshaiwi and Harries, 2021). Thus, it is therefore essential to explore this phenomenon not only by focusing on motivations but also by addressing barriers in order to minimize the gap between consumers' attitudes and behaviours toward SF.

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Thirdly, and in relation to the previous gap, despite its current relevance, research on SF commercialization in the HoReCa sector is minimal (Pandey *et al.*, 2023), and existing studies mainly focus on retail contexts (Cakar, 2022), which leaves a gap of how consumer behaviours differ across the retail and HoReCa sectors. This distinction is relevant since consumption drivers, barriers and motivations across industries may differ. For example, the retail sector offers consumers greater autonomy in decision-making, where intentions are often driven by economic considerations, perceived control, and personal motivations (Berri and Toma, 2023; Bolos *et al.*, 2022). The HoReCa sector presents a more constrained purchasing environment, where consumer choices depend on restaurant reputation, menu offerings, and the discretion of service providers (Moriuchi and Moriuchi, 2024; Pandey *et al.*, 2023). This gap hinders the development of effective strategies with a large potential for promoting SF consumption in these sectors in China.

To address these gaps, this study explores (1) how consumer attitudes and purchase intentions toward SF are formed in China; (2) the role of perceived risks in this process; and (3) how these dynamics differ between the retail and HoReCa sectors. To reach these goals, it adopts a dual-theory approach, integrating the theory of planned behaviour (here on TPB) and the attitude-behaviour-context (here on ABC) framework to analyse SF purchasing behaviour comprehensively. TPB, on the one hand, explains individual cognitive and motivational factors, including behavioural, normative, and controlling beliefs (Ajzen, 1991). The ABC framework, on the other hand, expands this perspective by incorporating external influences, such as economic factors, environmental considerations, and perceived risks, which shape consumer decision-making (Aral and López-Sintas, 2023; Sadiq *et al.*, 2023). This approach is particularly suited to China's context, where cultural norms, market dynamics, and sustainability trends intersect to shape consumer behaviour.

This research contributes to the literature by advancing theoretical understanding of SF purchasing behaviour. To the best of our knowledge, it is the first study to integrate cognitive, contextual, and cultural influences within the TPB-ABC framework to explain the food waste phenomenon. From a practical perspective, the findings provide sector-based actionable insights for businesses and policymakers to reduce perceived risks, improve SF accessibility, and build consumer trust in China. The results indicate that context-based strategies are required to promote SF consumption, with a special emphasis on reducing food safety and quality risk perceptions, and that strategies promoting consumers' perceived behavioural control can be effective in the retail sector, but not in HoReCa.

2. Theoretical framework and hypothesis development

2.1 The theory of planned behaviour (TPB)

This study combines TPB with the ABC model to explain Chinese consumer behavioural intentions and attitudes toward SF in the HoReCa and retail sectors. In this sense, TPB analyses the primary behavioural intention drivers toward SF, while the ABC model addresses contextual motivators and inhibitors impacting attitudes.

TPB is a robust model for predicting behavioural intentions, positing that attitude (individual's positive or negative evaluation to perform a behaviour, here on ATT), subjective norms (perceived social expectations from significant others that influence a person's behaviour and their willingness to comply with those expectations), and perceived behavioural control (individual's perception of their ability to perform such behaviour, considering factors like perceived difficulty, resources, and external barriers, here on PBC) collectively shape intentions, which subsequently drive actions (Ajzen, 1991). TPB has been widely applied in pro-environmental research, including organic food adoption (e.g. Cao *et al.*, 2024) and sustainable practices (e.g. Martini *et al.*, 2024) in different cultural contexts (e.g. Randall *et al.*, 2024). However, its application to SF consumption unveils significant limitations and challenges in culturally and contextually complex settings like China.

First, TPB overemphasizes social pressures in contexts where intrinsic motivations play a dominant role. Research on food waste behaviour suggests that subjective norms become less

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predictive when decisions are driven by internalized values rather than external influences (Filimonau *et al.*, 2023). Specifically, in sustainable food choices, individuals may act based on personal ethical convictions rather than perceived societal expectations (Chen, 2025; Prisco *et al.*, 2025). For example, in the context of organic food consumption, Testa *et al.* (2019) argued that subjective norms it did not impact consumers' intentions to purchase and a recent study by Chen (2025) found similar results in suboptimal food consumption. Consequently, this study, uses personal norms (here on PN) instead of subjective norms as a potential exogenous variable explaining individuals' intentions to purchase SF. This inclusion also answers the call from Randall *et al*. (2024) to investigate the role of PN within TPB in non-Western cultural context. PN refers to internalized moral obligations shaped by collectivist values (e.g. *mianzi*), where SF consumption aligns with social harmony and environmental duty (Al Mamun *et al*., 2024). PN has been found to be a strong predictor of behaviours rooted in ethical and environmental responsibility, particularly in collectivistic cultures like China, where moral duty and pro-social behaviour drive sustainable consumption (Al Mamun *et al*., 2024). Batool *et al.* (2024) highlighted that PN mediates the effects of altruistic, biospheric, and egoistic values on pro-environmental behaviour, further supporting the inclusion of PN in the TPB framework for predicting sustainable consumption behaviours. This approach also aligns with recent extensions of TPB in developing economies, such as Qalati *et al.* (2022), who demonstrated that PN significantly mediates energy-saving intentions in households, emphasizing the role of intrinsic values in collectivistic societies.

Second, given that the consumption of SF can occur through different distribution channels or sectors, the role of ATT, PN and PBC in explaining behavioural intentions towards this type of food should be further explored. Therefore, it is essential to identify if purchase intention (consumer's willingness and likelihood to buy a product or service, here on PI) differ based on the purchasing context, particularly between self-directed retail environments and externally controlled HoReCa settings. In retail, SF often involves uncooked, cooked, or semi-elaborated products that typically arise from overstocked inventory, including products that are surplus or nearing expiration, also referred to as suboptimal food (Aschemann-Witzel *et al.*, 2020). In the HoReCa sector, SF involves cooked meals that emerge from daily operations and encompasses excess food prepared for immediate consumption or unsold dishes (Pandey *et al.*, 2023). In retail settings, consumers assess SF products independently based on price, quality, and expiration date, often relying on habitual shopping patterns and heuristics rather than ethical considerations. Economic incentives including discounts and promotions play a dominant role in influencing purchase decisions (Aschemann-Witzel *et al.*, 2020; Berri and Toma, 2023). Consumers in retail environments also experience greater control over product selection, as they can inspect SF items directly before purchasing. In contrast, consumers in the HoReCa sector rely more on restaurant reputation, service trust, and menu offerings when making purchase decisions. Unlike retail shoppers, they do not select SF items directly but depend on the restaurant's commitment to food quality and sustainability. As a result, social perceptions, trust in the establishment, and perceived food safety have a stronger influence on SF consumption in HoReCa than in retail settings (Pandey *et al.*, 2023). This study proposes that, according to the principles of the TPB, ATT toward SF, PBC, and PN significantly explain consumers' PI for SF in both the retail and HoReCa sectors. However, the consideration and inclusion of these two distinct settings enhances understanding of how behavioural predictors effects vary depending on sector-specific conditions. Based on the above discussion, the following hypotheses are proposed:

H1. ATT toward SF positively influences consumers' intention to buy SF in the retail (H1a) and HoReCa (H1b) sectors.

H2. PBC over SF positively influences consumers' intention to buy SF in the retail (H2a) and HoReCa (H2b) sectors.

H3. PN regarding SF positively influences consumers' intention to buy SF in the retail (H3a) and HoReCa (H3b) sectors.

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2.2 Attitude-behaviour-context (ABC) theory

While TPB is a reliable model for predicting individuals' intentions based on cognitive aspects, its reliance on rational, deliberate evaluations overlooks the emotional and habitual nature of many consumer behaviours, particularly in culturally embedded contexts like food consumption (Martini *et al.*, 2024). For example, in rural China, dietary choices are often guided by tradition rather than calculated cost–benefit analyses, rendering TPB's rational framework inadequate to explain SF adoption (Niu *et al.*, 2025). Furthermore, TPB's narrow focus on individual cognition neglects external influences such as economic incentives, cultural norms, and policy interventions. Government measures like the "Clean Plate Initiative" reframe SF consumption as a civic duty, directly shaping behaviour beyond TPB's individualistic scope (Ogunmoroti *et al.*, 2022; Wang *et al.*, 2024). These limitations underscore the need for a complementary framework that accounts for contextual drivers and barriers.

Thus, this study integrates ABC theory, which posits that ATT alone cannot accurately predict behaviour, and that the context of ATT formation and behaviour is crucial for understanding behavioural formation (Aral and López-Sintas, 2023). Regarding SF consumption, purchasing decisions are shaped not only by cognitive evaluations but also by contextual factors including pricing strategies, accessibility, cultural norms (e.g. social perceptions of thriftiness), infrastructure (e.g. urban vs. rural distribution), and policy frameworks (Campos *et al.*, 2022; Cao *et al.*, 2023; Thanki *et al.*, 2024). China's collectivist culture and sustainability ethics uniquely shape consumer behaviour toward SF. The concept of *mianzi* (social reputation) creates a duality: while frugality may conflict with social status, SF consumption is increasingly framed as a civic duty under campaigns like the Anti-Food Waste Law (Wu *et al.*, 2023; Wang *et al.*, 2024). For instance, Liang *et al.* (2024) found that Chinese consumers prioritize environmental benefits over promotional appeal, which increases purchases of discounted SF. Similarly, traditional values align SF adoption with collective responsibility, mitigating perceived risks (Birkin *et al.*, 2021). ABC theory captures these nuances by incorporating economic, personal, and environmental drivers and contextual inhibitors to enhance understanding of SF consumption. This approach is effective for studying pro-environmental behaviours (e.g. Zhang *et al.*, 2021; Sadiq *et al.*, 2023). The dual framework captures China's unique interplay of intrinsic values and policy-driven sustainability (Aral and López-Sintas, 2023). Thus, this study employs attitudinal drivers including price consciousness, sales proneness, and food waste awareness to investigate their impacts on consumers' perception of SF. In addition, it also explores the role of perceived risks associated with SF as antecedents of consumers' ATTs towards the purchase of SF.

2.2.1 Price consciousness (PC) and sale proneness (SP). Regarding economic drivers, SF products are often priced lower than conventional foods to incentivize quick sales and mitigate food waste, making them attractive to consumers with strong price sensitivity (Hong *et al.*, 2023; Pandey *et al.*, 2023). Price consciousness (here on PC), defined as the willingness to seek lower prices and maximize value for money, plays a crucial role in shaping consumer ATTs towards SF (Kayikci *et al.*, 2022). Price-conscious consumers tend to allocate their budgets carefully and exhibit a preference for cost-effective purchasing decisions, leading them to favour SF products as a means of optimizing economic utility (Zhou and Ali, 2024). Similarly, sales proneness (here on SP), the tendency to prioritize discounted or promotional items, reinforces positive ATTs toward SF, as consumers who actively seek deals associate discounts with financial prudence and effective resource allocation (Yazdanparast and Kukar-Kinney, 2023). These consumers derive psychological gratification from securing bargains, which enhances their perception of SF as an economically rational and desirable option (Adel *et al.*, 2022). Therefore, it is proposed that:

H4. Consumers' PC positively influences their ATT toward SF consumption.

H5. Consumers' SP positively influences their ATT toward SF consumption.

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2.2.2 Food waste awareness (FWA). Personal and environmental factors have also been identified as drivers of pro-environmental behaviour (Randall *et al.*, 2024). In this sense, food waste awareness (here on FWA), which involves an individual's understanding and recognition of food waste issues, encompassing knowledge, ATTs, beliefs, and behaviours related to its reduction, prevention, and management, further affects consumer ATTs by linking SF consumption to broader sustainability goals (Schultz, 2001). Consumers with high FWA experience guilt over discarding edible food and recognize SF as a viable means to mitigate environmental harm (Zhang *et al.*, 2023). This ethical alignment is particularly salient in China, where national campaigns have heightened public awareness of food waste's environmental impact (Wang *et al.*, 2024). However, low FWA remains a barrier to SF adoption, as seen in studies where consumers dismissed suboptimal foods because of misconceptions about quality or safety (Cicatiello *et al.*, 2019; Chen, 2025). Previous studies highlighted that targeted communication strategies, such as emphasizing economic savings alongside environmental benefits, can amplify SF acceptability by enhancing perceived value (Varese *et al.*, 2023). Consequently, it is expected that:

H6. Consumers' FWA positively influences their ATT toward SF consumption.

2.2.3 Perceived risks. Perceived risks associated with SF, however, counteract these motivators. Previous studies have found that consumers associate SF with physical risks (health-related concerns), performance risks (quality-related concerns), psychological risks (negative impact on self-image), social risks (stigma linked to frugality), and financial risks (perception of poor value-for-money or financial loss) (Berri and Toma, 2023; Adel *et al.*, 2021). First, consumers associate SF with potential health risks, fearing contamination or spoilage due to shorter shelf life. For example, Sielicka-Różyńska and Samotyja (2022) found that misinterpretations of food labels exacerbate safety concerns, leading many consumers to reject SF. Second, consumers may doubt the quality of SF products regarding taste, freshness, and nutritional value, which heightens the perceived performance risk (Bolos *et al.*, 2022). For instance, Neubig *et al.* (2022) suggested that doubts about quality and safety in aged products reduce purchase likelihood. Third, owing to suboptimal attributes, SF consumption may negatively affect self-perception. For example, Grewal *et al.* (2019) suggested that choosing unattractive food affects self-signalling, leading to negative self-perception. Fourth, SF consumption may trigger embarrassment if linked to financial constraints or low social status (Hartmann *et al.*, 2021). Research shows that these stereotypes are particularly strong in collectivist societies like China, where social perception significantly influences PI (Liang *et al.*, 2024). Finally, consumers may fear economic loss if SF spoils before consumption. For example, Adel *et al.* (2021) found that financial concern is heightened for consumption of suboptimal food in China. Rising food prices exacerbate this, making consumers hesitant to purchase discounted surplus items if they doubt their usability (Grunert *et al.*, 2023). Thus, it is expected that stronger perceptions about the physical, performance, psychological, social, and financial risks of SF will reduce consumers' ATT toward it:

H7. Physical risk negatively influences ATT toward SF consumption.

H8. Performance risk negatively influences ATT toward SF consumption.

H9. Psychological risk negatively influences ATT toward SF consumption.

H10. Social risk negatively influences ATT toward SF consumption.

H11. Financial risk negatively influences ATT toward SF consumption.

2.2.4 Demographic variables. In line with prior sustainable consumer behaviour studies (Wani *et al.*, 2022; Qalati *et al.*, 2022; Cao *et al.*, 2024), as control variables, this study includes the socio-demographic factors age, gender, employment, income, education, and city size. For instance, gender and age were found to be relevant factors affecting sustainable behaviour during the COVID-19 lockdown in China (Li *et al.*, 2021). Younger consumers and women

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often show stronger pro-environmental tendencies (Li *et al.*, 2022), while higher income and education levels correlate with greater sustainability awareness (Asad, 2024; Hassan *et al.*, 2022). City size impacts consumer behaviour, as seen in studies conducted in cities like Beijing, Guangzhou, and Chongqing, where socio-demographic distributions influenced the adoption of sustainable shellfish (Wang and Somogyi, 2019). Thus, the inclusion of these variables will enhance the model's robustness. Figure 1 illustrates the research model.

3. Research methods

3.1 Data collection

This study was conducted in China, where food waste is a critical issue due to a large population, resource constraints, and rising food demand. Understanding consumer behaviour toward SF in this context is crucial, as China's policies and cultural factors influence ATT toward food waste and sustainable consumption (Xue *et al.*, 2021; Wang *et al.*, 2024; Niu *et al.*, 2025). The data were collected through an online survey distributed via WenJuanXing, a widely used Chinese panel company, employing random sampling to ensure a diverse sample of consumers responsible for food purchasing. The inclusion criteria required respondents to be at least 18 years old, actively involved in household food purchasing, and residing in China. This approach guarantees the representativeness of the sample in reflecting real consumer behaviours related to SF consumption. The survey was translated and back-translated for linguistic and cultural accuracy (Brislin, 1970).

A total of 695 valid responses were collected, surpassing the minimum required sample size of 652 determined through statistical power analysis. The power analysis was conducted using G*Power 3.1, assuming an effect size of 0.02 (small), a power level of 0.80, a significance level of 0.05 (95% confidence interval), and 14 predictors (Faul *et al.*, 2007). Sample demographics show that the majority of respondents were female (57.27%), aged 31–49 (45.12%), with graduate-level education (84.17%). Employment levels were high (91.65% full-time employed), with 89.5% identifying as omnivores. Income distribution showed that 32.81% had far above average earnings, and 22.3% resided in large cities (population above 10,000,000). A detailed description of the sample is presented in the Appendix.

3.2 Survey design and measurement

The survey utilized a randomized block design, ensuring that items within each construct were presented at random to different participants. This method reduces order bias, minimizes response fatigue, and enhances response validity (Podsakoff *et al.*, 2024). A seven-point Likert scale (1 = "strongly disagree" to 7 = "strongly agree") was used for all items, as it provides greater differentiation in responses than five-point scales, improving data precision and reliability (Allen *et al.*, 2022). All items have been validated in studies on consumer behaviour in China, ensuring cultural relevance and measurement reliability. A detailed description of the measurement items is presented in the Appendix. All the latent constructs were modelled as composites compatible with reflective measurement models (mode A). This measurement model was selected, since the items represent consequences of the construct. This means that a change in the construct (trait) implies a similar change in the indicators, and a high correlation among the indicators for the different constructs is expected. Thus, this approach provides robust measurement of consumer perceptions and behavioural constructs (Cepeda-Carrion *et al.*, 2019).

3.3 Common method bias assessment

Several procedural and statistical measures were implemented to mitigate common method bias and ensure data validity. To minimize potential biases arising from self-reported survey data, participants were assured of anonymity and confidentiality, reducing the likelihood of

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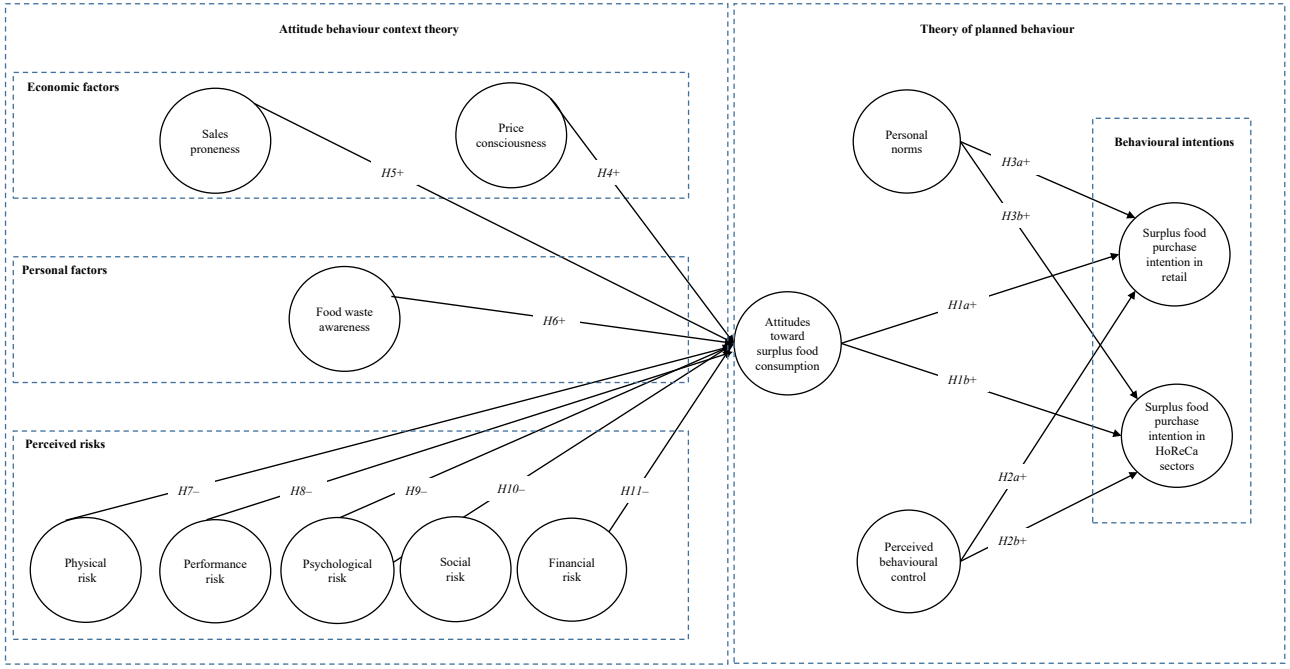


Figure 1. The conceptual framework. Source: Authors' own work

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social desirability bias. The questionnaire design incorporated randomization of items within blocks to prevent response patterns driven by order effects. Additionally, psychological separation techniques were applied (placing predictor and criterion variables on different pages), reducing the risk of respondents linking related constructs and adjusting their answers accordingly (Podsakoff *et al.*, 2024). This separation involved randomizing not just different blocks but also the order of presentation of constructs within blocks. Definitions of SF were provided to ensure a consistent understanding of key concepts, and survey items were phrased clearly to avoid ambiguity or leading responses. Harman's single-factor test indicated the first factor accounted for 31.7% of the variance, below the 40% threshold, with other eigenvalues evenly distributed (Baumgartner and Weijters, 2021). These analyses suggest common method bias was not a significant problem.

4. Results

This study aims to explain consumers' intention to purchase SF, integrating multiple theoretical perspectives into a composite reflective model. The model included numerous latent and manifest variables, uncovering both direct and indirect connections within a composite reflective structure. Therefore, we employed partial least squares structural equation modelling (statistical technique used to analyse complex relationships between variables, here on PLS-SEM) using SmartPLS v.4. software. PLS-SEM is preferred for exploratory research, complex predictive models with multiple constructs, and smaller sample sizes and is thus suitable here given the novelty of SF commercialization in China and the need to analyse sector-specific dynamics (Hair and Alamer, 2022).

4.1 Measurement model

To evaluate the measurement model, reliability, convergent validity, and discriminant validity were examined. Reliability was confirmed, as all outer loadings exceeded 0.605, and the composite reliability index (Rho_C) ranged from 0.825–0.931, surpassing the 0.70 threshold (Hair and Alamer, 2022). Convergent validity was established, as the average variance extracted for all constructs exceeded 50%, confirming that indicators adequately explained the variance in their respective constructs (Hair *et al.*, 2019). Discriminant validity was assessed using the heterotrait–monotrait ratio. All values fell below the conservative threshold of 0.85, with an acceptable limit of 0.90; thus, the constructs measured distinct theoretical concepts (Henseler *et al.*, 2014). Table 1 presents the outer loadings, Rho_C, average variance extracted, and variance inflation factor values, while Table 2 presents the heterotrait–monotrait ratios and variable correlations. Variance inflation factor quantifies how much the variance of a regression coefficient (beta) is inflated due to multicollinearity. If an independent variable is highly correlated with other independent variables, its variance inflation factor value will be high, meaning its standard error is larger than it should be. In PLS-SEM, variance inflation factor values above 3.3 suggest high collinearity, potentially distorting regression estimates. In this study, all variance inflation factor values were below 3.3, indicating that collinearity was not a concern and ensuring robust parameter estimation (Hair *et al.*, 2019).

4.2 Structural model

The significance of the structural factors was determined using a two-tailed bootstrapping test with 8,000 subsamples. Statistical significance was assessed by inspecting *p*-values. A *p*-value below 0.05, which is the typical threshold accepted in social sciences, suggests a significant effect at the 95% confidence level, while a *p*-value below 0.001 indicates very strong statistical significance, reducing the likelihood of error (Hair *et al.*, 2019). The model's explanatory power was examined through R^2 values, which indicate the proportion of variance explained in the dependent variables (Hair and Alamer, 2022). The model explained 64.1% of the variance of PIs in the retail sector and 54.8% in the HoReCa sector, demonstrating strong predictive

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Table 1. Measurement model

Construct/Indicator	Outer loadings	Variance inflation factor	Composite reliability index (Rho-C)	Average variance extracted
ATT		*2.69*	*0.936*	*0.830*
ATT1	0.917	–		
ATT2	0.909			
ATT3	0.908			
PBC		*1.627*	*0.825*	*0.616*
PBC1	0.875	–		
PBC2	0.847			
PBC3	0.605			
PN		*2.189*	*0.944*	*0.850*
PN1	0.929	–		
PN2	0.925			
PN3	0.912			
FWA		*1.202*	*0.843*	*0.641*
FWA1	0.952	–		
FWA2	0.934			
FWA3	0.937			
SP		*1.512*	*0.869*	*0.768*
SP1	0.881	–		
SP2	0.871			
PC		*1.549*	*0.908*	*0.832*
PC1	0.907	–		
PC2	0.917			
Performance risk		*3.114*	*0.935*	*0.828*
Performance risk 1	0.913	–		
Performance risk 2	0.915			
Performance risk 3	0.902			
Psychological risk		*1.725*	*0.867*	*0.767*
Psychological risk 1	0.953	–		
Psychological risk 2	0.790			
Social risk		*2.171*	*0.903*	*0.756*
Social risk 1	0.909	–		
Social risk 2	0.855			
Social risk 3	0.844			
Financial risk		*2.525*	*0.940*	*0.886*
Financial risk 1	0.945	–		
Financial risk 2	0.938			
Physical risk		*3.171*	*0.929*	*0.814*
Physical risk 1	0.921	–		
Physical risk 2	0.935			
Physical risk 3	0.848			
PI in retail		–	*0.932*	*0.821*
PI in retail 1	0.894			
PI in retail 2	0.908	–		
PI in retail 3	0.916			
PI in HoReCa sectors		–	*0.947*	*0.855*
PI in HoReCa sectors 1	0.924			
PI in HoReCa sectors 2	0.927	–		
PI in HoReCa sectors 3	0.923			

Source(s): Authors' own work

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Table 2. Discriminant validity analysis results

	1	2	3	4	5	6	7	8	9	10	11	12	13
1. ATT		0.261	−0.365	0.611	0.365	0.698	0.772	0.725	−0.449	−0.305	−0.43	0.314	−0.298
2. FWA	0.317		−0.268	0.327	0.22	0.235	0.287	0.264	−0.168	−0.161	−0.117	0.193	−0.24
3. Financial risk	0.412	0.34		−0.434	−0.082	−0.222	−0.386	−0.195	0.658	0.529	0.69	−0.017	0.654
4. PBC	0.74	0.457	0.561		0.278	0.465	0.588	0.464	−0.366	−0.218	−0.383	0.194	−0.298
5. PC	0.431	0.284	0.098	0.383		0.331	0.337	0.285	−0.07	−0.153	−0.06	0.563	−0.083
6. PI in HoReCa	0.77	0.277	0.249	0.549	0.386		0.731	0.673	−0.368	−0.267	−0.319	0.264	−0.22
7. PI in retail	0.863	0.352	0.437	0.727	0.399	0.809		0.666	−0.415	−0.285	−0.397	0.288	−0.313
8. PN	0.801	0.317	0.218	0.534	0.334	0.736	0.738		−0.362	−0.279	−0.317	0.214	−0.189
9. Performance risk	0.5	0.206	0.744	0.443	0.082	0.406	0.465	0.401		0.543	0.789	0.011	0.622
10. Psychological risk	0.344	0.193	0.627	0.261	0.173	0.302	0.318	0.321	0.637		0.525	−0.029	0.565
11. Physical risk	0.481	0.145	0.779	0.469	0.071	0.354	0.444	0.351	0.885	0.625		−0.005	0.582
12. SP	0.396	0.27	0.078	0.281	0.755	0.329	0.365	0.268	0.051	0.083	0.025		0.025
13. Social risk	0.342	0.317	0.769	0.401	0.1	0.251	0.364	0.212	0.718	0.676	0.674	0.066	

Note(s): Values above the diagonal indicate construct correlations. Values below the diagonal represent heterotrait–monotrait ratios

Source(s): Authors' own work

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capability. Additionally, 36.9% of the variance in ATT toward SF was explained, supporting the relevance of the proposed constructs in shaping consumer behaviour. Regarding model fit indices, the model standardized root mean square residual was 0.044 and the normed fit index was 0.813. Although the standardized root mean square residual is below 0.080, the low normed fit index could suggest low data–model fit. Nevertheless, typical goodness-of-fit indexes should be interpreted with caution in PLS-SEM because of its restrictive assumptions on the residual covariances (Lohmöller, 1989).

Table 3 presents the structural model estimates, including path coefficients, t-values, and R^2 values. The β coefficient indicates the strength and direction of the relationship between variables, where higher absolute values suggest stronger effects. A t-value greater than 1.96 corresponds to a *p*-value below 0.05, indicating statistical significance at the 95% confidence level (Hair *et al.*, 2019).

4.3 Hypothesis testing

The results provide empirical support for several theoretical propositions while also revealing unexpected findings that enhance understanding of consumer behaviour toward SF, as illustrated in Figure 2.

4.3.1 TPB predictors (H1–H3). As predicted in Hypothesis H1a, ATT toward SF significantly and positively influences PI in the retail sector ($\beta = 0.505$, $t = 11.984$, $p < 0.001$), while H1b confirms a similar effect in the HoReCa sector ($\beta = 0.416$, $t = 8.153$, $p < 0.001$). This strong positive effect aligns with TPB, suggesting that favourable ATT toward SF significantly influences consumers' PI (as found by Batool *et al.*, 2024; Cao *et al.*, 2024). As posited in Hypothesis H3a and H3b, PN positively predicts PI in both retail ($\beta = 0.177$, $t = 4.598$, $p < 0.001$) and HoReCa ($\beta = 0.350$, $t = 7.324$, $p < 0.001$). Thus, moral obligations and individual values are role for SF acceptance, in line with previous research in collectivist cultures (Chen, 2025; Al Mamun *et al.*, 2024). Similarly, PBC exhibits a positive and significant effect on PI in retail ($\beta = 0.177$, $t = 4.598$, $p < 0.001$), supporting hypothesis H2a. However, PBC does not significantly predict PI in HoReCa ($\beta = 0.047$, $t = 1.288$, $p = 0.198$), leading to the rejection of hypothesis H2b. This reveals differences between retail and HoReCa sectors, contextualized by the ABC framework. In retail, consumers exhibit greater autonomy, aligning with Cao *et al.* (2023), who found that retail context allows for deliberate, norm-

Table 3. Structural model results

Structural relationship	β	*t*-value	Hypothesis result
H1a. ATT → PI in retail	0.505	11.984***	Accepted
H1b. ATT → PI in HoReCa	0.416	8.153***	Accepted
H2a. PBC → PI in retail	0.177	4.598***	Accepted
H2b. PBC → PI in HoReCa	0.047	1.288	Rejected
H3a. PN → PI in retail	0.224	5.659***	Accepted
H3b. PN → PI in HoReCa	0.350	7.324***	Accepted
H4. PC → ATT	0.217	5.001***	Accepted
H5. SP → ATT	0.169	4.108***	Accepted
H6. FWA → ATT	0.107	2.951**	Accepted
H7. Physical risk → ATT	−0.187	3.189**	Accepted
H8. Performance risk → ATT	−0.260	4.275***	Accepted
H9. Psychological risk → ATT	−0.018	0.387	Rejected
H10. Social risk → ATT	0.045	0.837	Rejected
H11. Financial risk → ATT	−0.036	0.598	Rejected

R^2 – PI in retail = 0.641; R^2 – PI in HoReCa = 0.548; R^2 – ATT = 0.369

Note(s): ***$p < 0.001$, **$p < 0.01$; analysis performed at the 5% significance level

Source(s): Authors' own work

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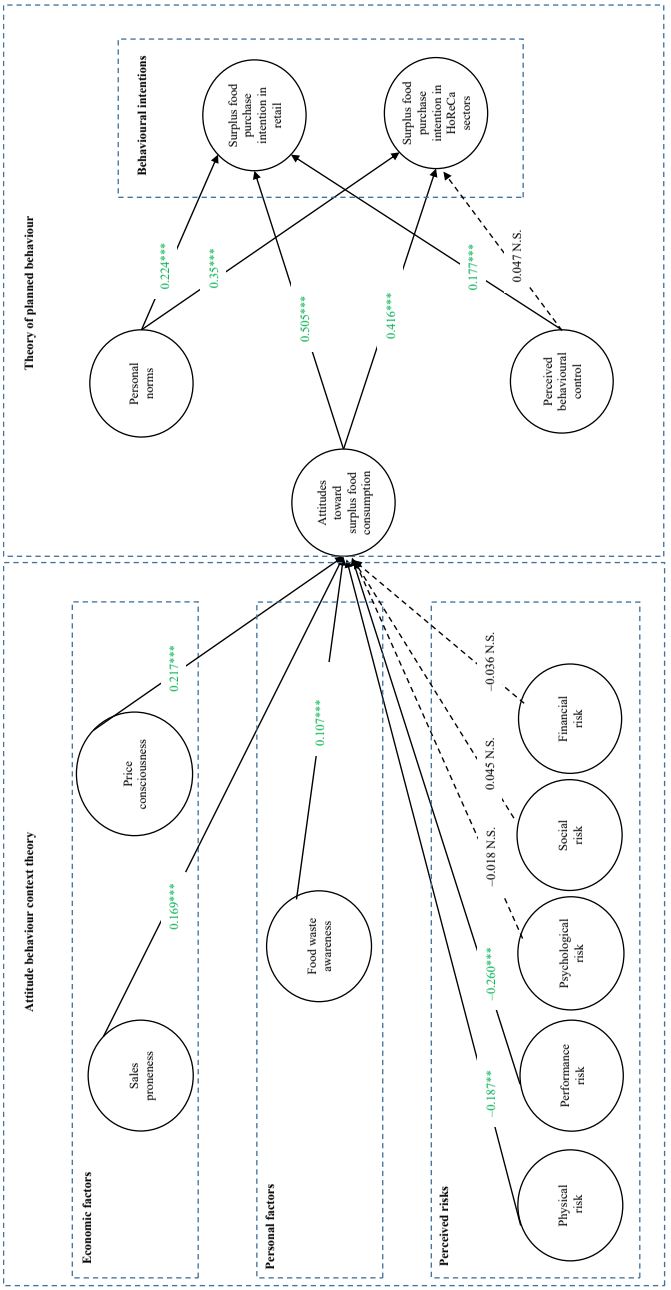


Figure 2. The structural model results. Source: Authors' own work

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driven behaviour. Conversely, in HoReCa, time constraints and menu rigidity limit autonomy, echoing Thanki *et al.* (2024)'s observation that "assessability" and "reachability" as main barriers for organic food adoption.

4.3.2 Motivational drivers (H4–H6). Regarding determinants of ATT, PC ($\beta = 0.217$, $t = 5.001$, $p < 0.001$) and SP ($\beta = 0.169$, $t = 4.108$, $p < 0.001$) positively influenced ATT, validating hypotheses H4 and H5. These findings align with previous studies, reflecting ABC theory's emphasis on economic context as a driver of behaviour, as discounted pricing strategies enhance the perceived value of SF (Adel *et al*., 2022). FWA also positively predicts ATT toward SF ($\beta = 0.107$, $t = 2.951$, $p = 0.003$), as predicted by hypothesis H6: heightened FWA enhances consumer acceptance of SF and reflects the growing influence of environmental consciousness on consumer behaviour as previous studies indicated (Zhang *et al*., 2023; Varese *et al*., 2023).

4.3.3 Risk barriers (H7–H11). Among risk factors, consistent with hypothesis H7 and H8, physical risk ($\beta = -0.187$, $t = 3.189$) and performance risk ($\beta = -0.26$, $t = 4.275$) impacted ATT negatively, underscoring ABC theory's focus on contextual inhibitors (Sadiq *et al*., 2023). These results highlight the importance of addressing safety and quality concerns to improve SF acceptability. In contrast, psychological risk ($\beta = -0.018$, $t = 0.387$, $p = 0.699$), social risk ($\beta = 0.045$, $t = 0.837$, $p = 0.403$), and financial risk ($\beta = -0.036$, $t = 0.598$, $p = 0.550$) do not significantly influence ATT, leading to the rejection of H9–H11. The lack of significance for psychological risk contradicts findings by Adel *et al*. (2021), suggesting that SF purchasing is no longer perceived as a reflection of personal self-image, likely due to increased normalization of sustainable consumption behaviours and environmental awareness. Similarly, the non-significance of social risk indicates that concerns about external judgement from peers or broader societal norms are not primary determinants of ATT, highlighting the potential for shifting consumer perceptions around SF acceptance. This aligns with China's collectivist ethos, where measures like the Anti-Food Waste Law normalize SF consumption as socially responsible, reducing stigma (Wang *et al*., 2024).

The non-significant influence of financial risk on consumers' ATT also contradicts previous findings (Hartmann *et al*., 2021; Adel *et al*., 2021). This might be explained by the stability of food prices in China versus Western countries (Grunert *et al*., 2023). While consumers traditionally weigh financial concerns heavily in food purchasing decisions, the consistently low pricing of SF may mitigate these apprehensions (Choi *et al*., 2022). Additionally, individuals already inclined toward SF consumption may exhibit higher levels of sustainability awareness, making them less sensitive to financial considerations in favour of environmental benefits (Liang *et al*., 2024). Future research should investigate whether financial concerns are more pronounced among consumer segments with lower disposable incomes or in markets where food pricing structures are more variable. Nevertheless, we did not find significant influences of the control variables on the dependent variables.

5. Conclusions

5.1 Theoretical contributions

This study contributes to a better understanding of SF commercialization literature by exploring how consumer intentions to purchase SF in retail and HoReCa sectors are shaped by general SF beliefs, personal characteristics and perceived risks. To do so, the study proposes and tests an integrative model, based on the TPB, that replaces subjective norms with PNs, which better captures moral obligations in collectivist cultures, and integrates ABC to contextualize economic and policy drivers of food waste ATTs formation (Al Mamun *et al*., 2024; Aral and López-Sintas, 2023). The TPB components capture individual cognitive and motivational factors driving SF consumption, while the ABC perspective accounts for external contextual influences including economic and environmental factors and perceived risks. Unlike previous models, that emphasize financial or sustainability-related motives in specific purchasing contexts (Berry and Toma, 2023; Adel *et al*., 2022), this study's dual-framework

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approach highlights the sector-specific nuances that shape SF acceptance, recognizing the contrasting purchasing dynamics in the retail and HoReCa sectors. Finally, this research expands the literature on SF consumption in Asian cultures, providing a unique perspective by examining SF consumption in China. The results yield significant theoretical insights.

Regarding the role of the TPB factors, the results largely align with prior research (Batool *et al*., 2024), showing that general ATTs towards SF and PN concerning SF consumption explain the intention to buy SF in both settings. In general terms, the study's results confirm that ATTs and PNs positively influence Chinese consumers' intention to purchase SF. The identified effects suggest that the patterns of influence are similar across retail and HoReCa channels. These results are also consistent with earlier studies demonstrating the significance of PN in predicting pro-environmental behaviours (Cao *et al*., 2024; Prisco *et al*., 2025), thus emphasizing the relevance of integrating individuals' moral aspects into models that analyse socially desirable or sustainable behaviours. However, differences were also observed. Findings reveal that PBC increases consumers' PIs in the retail sector, but not in the HoReCa. This effect can be explained because of different accessibility and diversity of SF options for consumers across these sectors. In retail settings, SF products (e.g. near-expiry groceries, discounted bulk items) are often prominently displayed, allowing consumers to physically inspect, compare, and select items based on personal preferences. Previous studies suggest that greater accessibility and diversity of SF products in retail facilitate consumer autonomy in purchase decisions (Hartmann *et al*., 2021; Lyu *et al*., 2024). This autonomy enhances PBC, which aligns with the positive effect of PBC on PI in retail settings in our study. Conversely, HoReCa consumers have limited direct control over SF purchases (Derqui and Filimonau, 2024). In contrast, HoReCa SF offerings (e.g. surplus meals, blind box surplus offer) are typically limited to menu offerings. The non-significant effect of PBC in the HoReCa sector reflects this structural constraint, suggesting that consumers perceive less autonomy in their SF purchasing decisions. This aligns with ABC theory's emphasis on contextual barriers, where external factors (e.g. purchase context) impact the translation of ATT into behaviour (Aral and López-Sintas, 2023). Additionally, some particular companies, such as small and medium-sized enterprises in HoReCa, may lack the technological capabilities (Ta'Amnha *et al*., 2024), knowledge (Asad, 2024), or green innovation capabilities (Asad *et al*., 2024) to promote SF options effectively into the market.

As for the factors that explain consumers' ATT formation, the study's findings highlight that concerns over food safety and quality significantly shape ATT toward SF consumption in China. These results reflect how tangible product attributes influence decision-making beyond abstract norms (Hartmann *et al*., 2021). These findings also echo that consumers prioritize SF items with clear expiration labels or quality certifications, bridging the gap between intention and action (Sielicka-Różyńska and Samotyja, 2022). However, psychological, social, and financial risks do not significantly influence ATT, suggesting that consumers no longer associate SF consumption with personal stigma, financial burden, or social disapproval in contrast with previous studies (Adel *et al*., 2021). A possible explanation is that affordability perceptions reduce financial concerns, while increasing eco-consciousness fosters social responsibility, making SF purchases more socially acceptable (Ta'Amnha *et al*., 2024). For example, platforms like Hottmax in China emphasize "value for money" messaging, reframing SF as a savvy choice rather than a financial loss. In addition, China's Anti-Food Waste Law in 2021 and nationwide campaigns (e.g. "Clean Plate Initiative") have normalized SF consumption as a civic duty, aligning it with collectivist values like social harmony (Ogunmoroti *et al*., 2022; Wang *et al*., 2024). This policy-driven shift reduces social stigma, as SF purchases are now perceived as socially responsible rather than frugal. Growing environmental awareness, particularly among urban Chinese consumers, has redefined SF consumption as an ethical act. Choosing SF is increasingly tied to self-fulfilment, where individuals derive pride from contributing to sustainability goals (Wani *et al*., 2023). This psychological reward offsets concerns about self-image or social judgement (Liang *et al*., 2024). These findings suggest that SF strategies in China must prioritize economic incentives

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(e.g. dynamic pricing), policy alignment (e.g. labelling SF as "sustainable"), and cultural narratives (e.g. linking SF to national sustainability goals) to further mitigate residual risks.

The results of this study also suggest that economic incentives emerge as key drivers of ATT toward SF, reinforcing the affordability–consumer acceptance link. As for SP, consumers with a higher propensity for seeking bargains associate financial savings with value and are more likely to view SF as an opportunity to obtain products at lower prices (Yazdanparast and Kukar-Kinney, 2023). This relationship implies that consumers' desire to take advantage of discounts influences their ATTs towards SF as a cost-effective choice. This reinforces the idea that promotional offers related to SF can effectively attract a segment of the population that is driven by price incentives (Cakar, 2022; Adel *et al*., 2022). Similarly, PC also plays a crucial role in shaping positive ATTs toward SF. Price-conscious consumers are particularly motivated by the potential to save money while still acquiring quality products (Zhou and Ali, 2024). The positive impact of PC on SF consumption underscores the importance of pricing strategies in encouraging the adoption of sustainable food practices in China. Generally, these findings emphasize that economic motivations, particularly the desire for lower prices and better value, are central to consumer ATTs toward SF in China. In a culture where price sensitivity is high due to the competitive nature of the market and the value placed on economic efficiency, these findings suggest that Chinese consumers are particularly receptive to SF as a cost-effective and practical solution (Adel *et al.*, 2022). Additionally, with China's large and diverse consumer base, where citizens face rising living costs, SF consumption can be perceived as an opportunity to save money while making a sustainable choice (Chen, 2025).

Finally, findings suggest that FWA fosters positive ATT toward SF, demonstrating that increased FWA enhances consumer willingness to purchase SF. This is consistent with previous studies that suggest that FWA may link SF consumption to broader sustainability goals (Schultz, 2001), as individuals with heightened awareness are more likely to adopt environmentally responsible behaviours, driven by a sense of moral obligation and environmental concern. Furthermore, this finding highlights the growing recognition of sustainability issues among Chinese consumers incentivized by increasing national and governmental focus on food waste reduction campaigns such as the "Clean Plate Campaign" (Ogunmoroti *et al*., 2022; Wang *et al*., 2024). This cultural shift towards greater awareness of food waste aligns with the results of this study, indicating that Chinese consumers who are more conscious of the environmental and social impacts of food waste are more likely to have positive ATTs toward SF. The positive effect of FWA suggests that as consumers become more informed about the environmental consequences of waste, they are more inclined to view SF as a sustainable and responsible alternative (Zhang *et al*., 2023).

5.2 Managerial implications

This study's results also offer relevant conclusions for practitioners and policymakers. According to the findings, to improve Chinese consumers' attitudes towards SF, companies and policy makers should mainly focus their strategies on reducing consumers SF perceptions around the risks related to performance and health issues. For example, retailers could apply SF certification labels or programmes (e.g. Zero-Waste Certified) to highlight health and quality benefits, while HoReCa companies could apply strict quality control measures in their processes to ensure that offered SF meets high safety and freshness standards. Generally, speaking, marketing efforts could also apply personalized engagement strategies that highlight affordability, product quality, and sustainability benefits. Regarding the relevant role of economic drivers, for example, retailers could utilize artificial intelligence driven platforms to offer real-time discounts on SF products based on purchase history or product expiration date. Push notifications could also alert price-conscious shoppers to nearby SF deals, leveraging economic incentives (Kayikci *et al*., 2022). Companies in the HoReCa sector can also design promotional engaging strategies like rewarding diners who choose SF with points redeemable

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for discounts, aligning with collectivist values and ethical fulfilment (Gao *et al.*, 2021; Birkin *et al.*, 2021).

In the retail sector, loyalty programs, along with actions to promote consumers' PBC, can also contribute to increasing SF sales in this sector. For example, empowering consumers with updated information about the potential benefits and uses of SF (e.g. portion guides, recipes, sensory information, etc.) could increase their control perceptions about this food. Also, training retail employees to better educate customers on how to manage SF could be beneficial. As PBC does not affect intentions in the HoReCa sector, companies in this sector should concentrate their strategies on other strategies emphasizing ATTs and PNs. Businesses should consider fostering a sense of personal responsibility among consumers through awareness campaigns and educational initiatives. For instance, collaborating with eco-influencers and leveraging AI-driven platforms can convert consumer trust and autonomy into measurable sustainability impacts—such as conserving water, energy, and agricultural resources. By aligning economic, ethical, and regulatory priorities, the commercialization of SF can be effectively scaled to support a more sustainable and efficient global food system. While price remains a vital factor, it should be complemented by focusing on environmental and social benefits and aligning SF with sustainability and community support (Varese *et al.*, 2023). For example, encouraging customers to share their experiences with SF through user-generated content could increase consumers'' SF acceptance. Running social media contests or challenges where users showcase creative recipes using surplus ingredients can inspire others to adopt sustainable consumption practices.

As for policymakers, they could introduce incentives for businesses to collaborate with SF initiatives. This could include tax benefits, grants, or recognition programs for companies actively engaged in SF initiatives. Policymakers should also enforce quality assurance regulations, encompassing mandatory labelling and safety standards for SF products, which are crucial to ensure consumer health and reduce their quality concerns toward SF. To further address consumer concerns about food safety and performance risks, it is essential to standardize SF labelling and safety regulations. Additionally, policymakers should invest in public awareness campaigns emphasizing SF's role in reducing food waste and its environmental benefits. For example, educational initiatives promoting SF as a healthy, safe, and sustainable food alternative can help reframe consumers and counteract barriers, particularly among eco-conscious consumers. By aligning policies with the study's findings and focusing on affordability, quality, and sustainability, policymakers can help promote the growth of the SF market and stimulate positive consumer ATTs and intentions. These measures not only benefit consumers but also contribute to broader sustainability and food waste reduction goals.

5.3 Limitations and future research lines

This study presents certain limitations, which represent future research avenues. First, the reliance on self-reported survey data introduces potential biases, although procedural and statistical controls were applied to mitigate common method bias. While this bias seems not to be present in this study, it cannot be entirely discarded. Future studies should apply alternative techniques, like the test marker variable approach, to control for this bias. Second, the omission of potentially relevant variables (e.g. past experiences with SF, or exposure to sustainability campaigns) may limit the model's predictive accuracy. Third, the sample, while representative of Chinese food purchasers, was predominantly female. This may influence the generalizability of the findings, although it aligns with research indicating that female consumers in China exhibit greater awareness of food sustainability and are more likely to engage in eco-friendly purchasing behaviours (Li *et al.*, 2022). Nevertheless, this demographic skew could influence consumer ATT toward SF. Thus, future studies should explore potential gender differences in SF purchasing behaviour. Fourth, given the dynamic nature of consumer ATT toward SF, longitudinal studies are needed to track how perceptions shift over time,

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particularly in response to policy changes, economic fluctuations, and sustainability awareness campaigns. Fifth, the single-country sample limits generalizability. Future research should conduct cross-country comparisons, particularly in regions with contrasting food safety standards and sustainability initiatives (e.g. Europe vs. Southeast Asia), to investigate how cultural norms influence SF acceptance and risk perception. Studies have shown that food safety concerns, social stigma, and economic motivations vary significantly by culture (Campos *et al*., 2022; Randall *et al*., 2024). Additionally, government regulations, such as food labelling policies and anti-waste legislation, may impact consumer confidence in SF differently across countries (Aral and López-Sintas, 2023). Future studies should investigate how entrepreneurial networks (Satar *et al.*, 2024) and stakeholder collaboration (Kanaan *et al.*, 2024) enable knowledge sharing and innovation in SF supply chains, particularly in regions with fragmented regulatory frameworks. Examining how SF consumption patterns evolve in different regulatory environments will enhance understanding of global SF behaviour. Cross-cultural comparisons could test how policy frameworks (e.g. EU vs. China) moderate the role of economic incentives, while longitudinal designs might reveal how repeated exposure to SF campaigns reshapes psychological risk perceptions (Grunert *et al*., 2023). Finally, future studies should examine how external interventions, including educational initiatives, certification schemes, and retailer transparency efforts, reduce perceived risks and enhance consumer confidence in SF purchasing.

In conclusion, SF is becoming increasingly mainstream, driven by rising environmental consciousness, cost-saving incentives, and regulatory catalysts like China's Anti-Food Waste Law. The integration of SF into the retail and HoReCa sectors is crucial to reducing food waste and promoting sustainable consumption. The societal impacts of SF adoption extend beyond waste reduction. China's policy-driven reframing of SF as a civic duty (e.g. Clean Plate Initiative) has normalized ethical consumption, fostering collective responsibility for resource conservation (Wang *et al*., 2024). Economically, SF bridges affordability and accessibility, reduces food insecurity while curbing landfill emissions (Hong *et al*., 2023). Environmentally, every redirected surplus meal mitigates methane emissions, advancing sustainable development goals (e.g. responsible production and consumption). However, overcoming barriers (e.g. food safety and quality concerns) requires coordinated efforts across sectors.

Acknowledgments

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(The Appendix follows overleaf)

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Appendix

Table A1. Demographic profile of participants

Variable		Frequency	%
Gender	Male	295	42.45%
	Female	398	57.27%
	Other	2	0.29%
	Total	695	100%
Age	18–30	280	40.29%
	31–49	390	45.12%
	50–64	25	3.6%
	65+	0	0%
	Total	695	100%
Income	Far below average	37	5.32%
	Somewhat below average	39	5.61%
	Average	183	26.33%
	Somewhat above average	208	29.93%
	Far above average	228	32.81%
	Total	695	100%
Education	Primary school	1	0.14%
	secondary school	40	5.76%
	Graduate	585	84.17%
	Postgraduate	64	9.21%
	Prefer not to say	5	0.72%
	Total	695	100%
Employment	Full time	637	91.65%
	Part time	20	2.88%
	Unemployed looking for job	8	1.15%
	Unemployed not looking for job	0	0%
	Retired	2	0.29%
	Student	25	3.6%
	Others	3	0.43%
	Total	695	100%
Size of the city	Below 200,000	21	3.02%
	200.001–500,000	65	9.35%
	500,001–1,000,000	100	14.39%
	1,000,001–3,000,000	140	20.14%
	3,000,001–5,000,000	97	13.96%
	5,000,001–10,000,000	117	16.83%
	Above 10,000,000	155	22.30%
	Total	695	100%
Diet	Vegan	8	1.15%
	Vegetarian	34	4.89%
	Pescatarian	5	0.72%
	Omnivore	622	89.5%
	Others	26	3.74%
	Total	695	100%

Source(s): Authors' own work

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Table A2. Measurement scale items

Construct	Acronym	Definition	Measurement scales	Source
Attitude	ATT	Individual's positive or negative evaluation	(1) Buying surplus food is a good idea (2) Buying surplus food is a wise choice (3) I like the idea of buying surplus food	Adopted from (Ajzen, 1991; Wong *et al.*, 2018)
Perceived Behavior Control	PBC	Individual's perception of their ability to perform such behavior	(1) If I wanted to, I could buy surplus food (2) I think it is easy for me to buy surplus food (3) It is mostly up to me whether or not to buy surplus food	Adopted from (Ajzen, 1991; Wong *et al.*, 2018)
Personal Norms	PN	Internalized moral obligations shaped by collectivist values (e.g. mianzi), where SF consumption aligns with social harmony and environmental duty	(1) Due to my personal values, I feel obliged to purchase surplus food (2) No matter what others think, I feel personally obliged to buy surplus food (3) I feel a moral obligation to buy surplus food	Adopted from (Issock *et al.*, 2020)
Food Waste Awareness	FWA	Individual's understanding and recognition of food waste issues	(1) Food waste increases the burden on the environment (2) Food waste impacts planetary resources (3) Food waste affects undernourished people globally	Reduced scale adopted from (Puteri *et al.*, 2022)
Sales Proneness	SP	Tendency to prioritize discounted or promotional items	(1) I am more likely to buy products on sale (2) If a product is on sale, I consider it a good reason to buy	Reduced scale of Lichtenstein *et al.* (1993) adopted by (Adel *et al.*, 2022)
Price Consciousness	PC	Willingness to seek lower prices and value-for-money	(1) I am willing to make extra effort to find lower prices (2) The money saved by finding low prices is worth the time and effort	Reduced scale of Lichtenstein *et al.* (1993) adopted by (Adel *et al.*, 2022)
Social Risk	N/A	Stigma linked to frugality	(1) Choosing surplus food would not give me a sense of others' approval (2) Others do not react positively when I buy surplus food (3) Buying surplus food would harm my social image	Adopted from (Zeng, 2022)
Psychological Risk	N/A	Negative impact on self-image	(1) Buying surplus food would not boost my self-esteem. (2) Buying surplus food would not enhance my self-image	Adopted from (Zeng, 2022)

(*continued*)

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Table A2. Continued

Construct	Acronym	Definition	Measurement scales	Source
Performance Risk	N/A	Quality related concerns	(1) I distrust the quality of surplus food (2) I question the features (taste, freshness) of surplus food (3) Surplus food is not as good as conventional food	Adopted from (Zeng, 2022)
Financial Risk	N/A	Perception of poor value-for-money or financial loss	(1) I feel I would lose money's worth if I bought surplus food (2) Surplus food does not represent good value-for-money	Adopted from (Adel *et al.*, 2021)
Physical Risk	N/A	Health-related concerns	(1) I would put my health at risk by consuming surplus food (2) I would risk health side-effects by consuming surplus food (3) Conventional food is healthier than surplus food	Adopted from (Zeng, 2022)
Purchase Intention in retail	PI in retail	Likelihood of buying surplus food in retail settings (e.g. supermarkets)	(1) I intend to buy surplus food in supermarkets/ food stores (2) I will try to buy surplus food in supermarkets/ food stores (3) I plan to buy surplus food in supermarkets/ food stores	Adopted from (Ajzen, 1991, 2011)
Purchase Intention in HoReCa	PI in HoReCa	Likelihood of buying surplus food in hospitality settings (e.g. restaurants)	(1) I intend to buy surplus food in restaurants/ cafeterias (2) I will try to buy surplus food in restaurants/ cafeterias (3) I plan to buy surplus food in restaurants/ cafeterias	Adopted from (Ajzen, 1991, 2011)

Source(s): Authors' own work

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